

Vraj Patel

Systems Integration Engineer | Backend, Data, and Applied ML
Boulder, Colorado | [linkedin.com/in/vrajmp](https://www.linkedin.com/in/vrajmp)

PROFILE

Systems integration engineer building dependable backend services, operational data pipelines, and applied machine-learning systems at CU Boulder's Institute of Behavioral Science. Research background in large-scale public-interest datasets, recoverable long-running workflows, and satellite cybersecurity.

EDUCATION

University of Colorado Boulder, College of Engineering and Applied Science

M.S. in Computer Science and M.S. in Engineering Management - in progress

University of Colorado Boulder, College of Engineering and Applied Science

B.S. in Computer Science with a Business Minor - May 2025 | GPA 3.65/4.00 | Dean's List, 5 semesters

EXPERIENCE

Systems Integration Engineer

University of Colorado Boulder - Institute of Behavioral Science | May 2025 - Present | Boulder, CO

- Build an internal FastAPI and PostgreSQL ticket-intelligence platform with ML-assisted incident tagging and human review boundaries.
- Created a React, Vite, and Recharts dashboard that surfaced a Tuesday 10 a.m. ticket-volume peak for staffing analysis.
- Automated Active Directory group-membership cleanup with PowerShell, reducing an offboarding run from 15 minutes to under one minute.
- Managed Windows Autopilot and Jamf Pro enrollment for 50+ endpoints to meet Secure Compute requirements.

Research Assistant, Data Engineering and ML

Studio Lab - University of Colorado Boulder | March 2023 - May 2026 | Remote

- Built a cloud-hybrid pipeline for 40+ GB of 2014 and 2019 election speeches and Lok Sabha debates from 1981 to 2024.
- Implemented AWS S3, Transcribe, and Translate workflows with custom language models for political entities and lower transcription error.
- Built resumable Selenium crawlers with SQLite state and parsed thousands of PDFs into standardized ministry records.

Undergraduate Research Assistant - Satellite Telemetry Data Analysis

The Data Mine - Purdue University at L3Harris | August 2024 - May 2025 | West Lafayette, IN

- Developed anomaly-detection models for cyberattack patterns in NASA's NOS3 satellite telemetry and mapped scenarios to SPARTA and MITRE ATT&CK.
- Presented findings at the Data Mine of the Rockies Symposium to industry and government stakeholders.

Lead Technical Research Assistant

University of Colorado Boulder - Institute of Behavioral Science | March 2023 - May 2025 | Boulder, CO

- Built PyTorch and ResNet50 workflows for visual-bias analysis of New York Times COVID-19 imagery.
- Built resilient Selenium, BeautifulSoup, and LegiScan ingestion workflows with checkpoints, caching, and rate-limit handling for 15+ years of legislative data.

SELECTED PROJECTS

Operational Ticket Intelligence

FastAPI, PostgreSQL, React, Vite, Recharts, PyTorch

Internal platform combining dependable backend workflows, operational analytics, and ML-assisted incident tagging with human review boundaries.

Indian Parliamentary Data

Selenium, AWS, PyMuPDF, Data Engineering

Resumable pipelines for 40+ years of parliamentary debates, 40+ GB of campaign audio, and thousands of digitized documents.

SeeMyRace: Race Photo Search

Flask, Docker, PostgreSQL, DeepFace, PaddleOCR

Collaborative race-photo platform with selfie search, bib detection, moderation, and asynchronous image processing.

Mental Health NLP Research

PyTorch, Hugging Face, LLaMA 3.2, LoRA, TensorFlow

Academic NLP research using BERT classifiers and LLaMA fine-tuning on 53,000+ text samples; not a clinical product.

HateBlocker: Toxic-Content Detection

FastAPI, BERT, XGBoost, TF-IDF, Chrome Extension

Browser extension and API that detect and hide toxic content; BERT reached 91.5% accuracy on about 24,800 labeled tweets.

Self-Hosted Home Lab

TrueNAS, Docker Swarm, Portainer, Cloudflare Tunnel, Tailscale

Operate a 24 TB home server with containerized media and photo services plus authenticated remote access.

RESEARCH WRITING

Developing a paper on secure GitHub repository collaboration for work governed by federal data-protection agreements.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PowerShell, Bash

Systems: FastAPI, PostgreSQL, React, Docker, AWS, Git, Linux

ML and Data: PyTorch, Hugging Face, Pandas, OpenCV, Selenium, BeautifulSoup